

Infinite series

Math 163 • Prof. Chowdhury

The next page has 15 ideas as boxes, with 16 numbered arrows between them and nothing written on any arrow. An arrow is a relationship between two ideas, and not always an implication. Sometimes the first box gives you the second. Sometimes it doesn't, and there's a counterexample. Sometimes each gives the other. And sometimes it does give it, but on hypotheses you wouldn't have guessed.

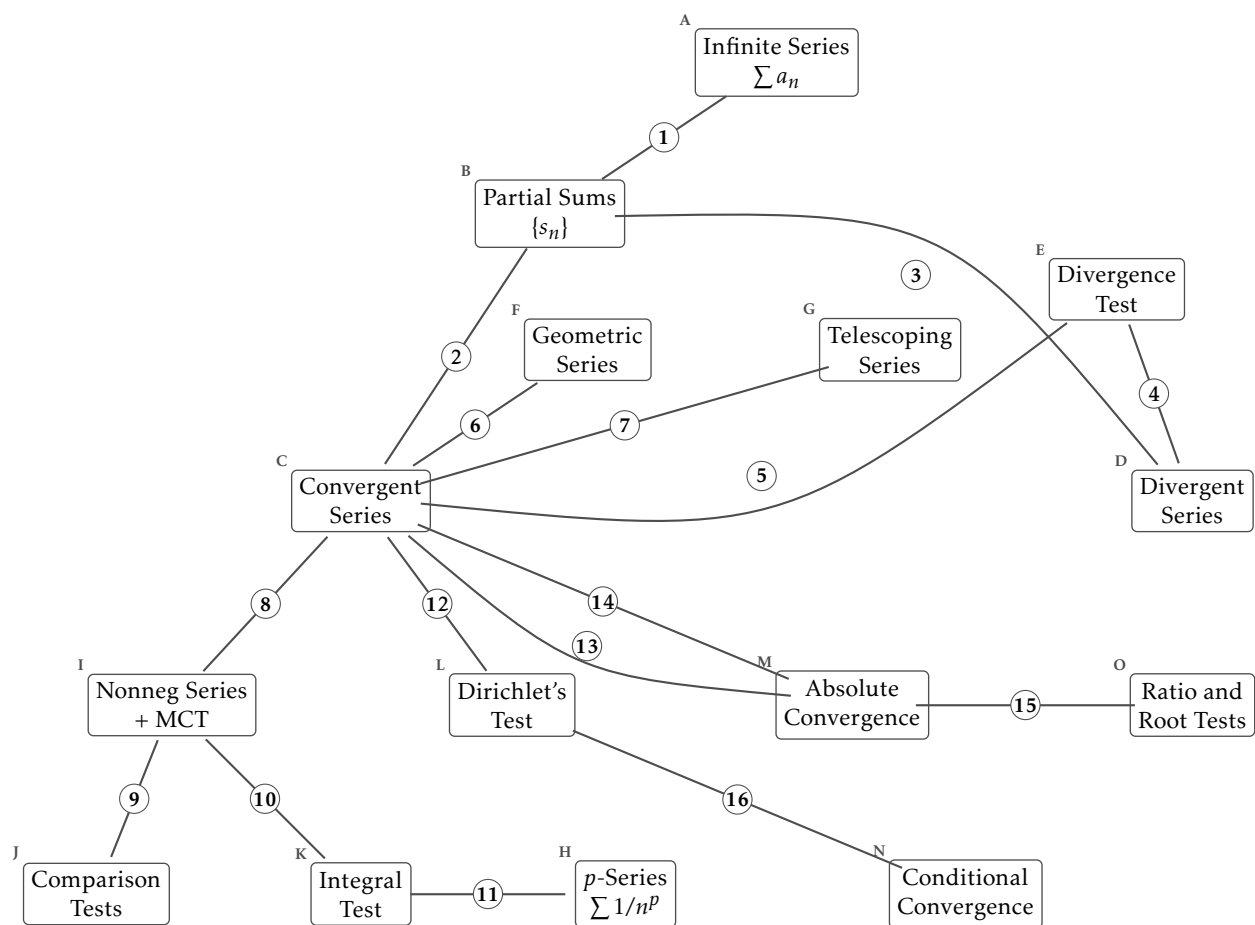
For each arrow, circle which of those four it is, and describe the relationship in one sentence. Name the theorem where there is one, and say what goes wrong where there isn't.

What each box means

- A. **Infinite Series** $\sum a_n$. The expression $a_1 + a_2 + a_3 + \dots$. It has no meaning until the partial sums give it one.
- B. **Partial Sums** $\{s_n\}$. $s_n = a_1 + \dots + a_n$. An ordinary sequence, so everything you know about sequences applies to it.
- C. **Convergent Series**. $\sum a_n$ converges when $\{s_n\}$ has a finite limit, and the sum is that limit.
- D. **Divergent Series**. $\sum a_n$ diverges when $\{s_n\}$ fails to converge, whether it runs away or oscillates.
- E. **Divergence Test**. If $a_n \not\rightarrow 0$ then $\sum a_n$ diverges. It is a test for divergence only, and it never certifies convergence.
- F. **Geometric Series**. $\sum ar^n$, which converges exactly when $|r| < 1$, with sum $a/(1-r)$ and partial sums $a(1-r^n)/(1-r)$.
- G. **Telescoping Series**. $\sum(b_n - b_{n+1})$, whose partial sums collapse to $b_1 - b_{n+1}$.
- H. **p-Series** $\sum 1/n^p$. Converges exactly when $p > 1$. The boundary $p = 1$ is the harmonic series, which diverges.
- I. **Nonneg Series + MCT**. If $a_n \geq 0$ the partial sums increase, so the monotone convergence theorem turns convergence into boundedness.
- J. **Comparison Tests**. Direct: $0 \leq a_n \leq cb_n$ eventually and $\sum b_n$ convergent gives $\sum a_n$ convergent. Limit: $\lim a_n/b_n \in (0, \infty)$ gives the same behaviour for both.
- K. **Integral Test**. If f is positive, continuous and eventually decreasing with $f(n) = a_n$, then $\sum a_n$ and $\int_1^\infty f$ converge or diverge together.
- L. **Dirichlet's Test**. If a_n decreases to 0 and the partial sums of b_n are bounded, then $\sum a_n b_n$ converges. Proved by summation by parts.
- M. **Absolute Convergence**. $\sum a_n$ converges absolutely when $\sum |a_n|$ converges. This is strictly stronger than convergence.
- N. **Conditional Convergence**. $\sum a_n$ converges conditionally when it converges and $\sum |a_n|$ does not. The convergence rests on cancellation.
- O. **Ratio and Root Tests**. With $L = \lim |a_{n+1}/a_n|$ or $L = \limsup |a_n|^{1/n}$: absolutely convergent if $L < 1$, divergent if $L > 1$, and no conclusion if $L = 1$.

Counterexamples and benchmarks

- The harmonic series $\sum 1/n$ diverges even though its terms tend to zero.
- The alternating harmonic series $\sum (-1)^{n+1}/n$ converges conditionally, to $\ln 2$.
- $\sum \sin(n\theta)/n$ converges by Dirichlet's test and is not alternating.
- Summation by parts is the discrete integration by parts, and it is what proves Dirichlet's test.
- The Cauchy criterion for series: control every far enough tail at once, not one chosen tail.



Every arrow is drawn the same way on purpose. Deciding the kind is half of each answer.

The arrows

1. gives / fails / both ways / gives, other hypotheses

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2. gives / fails / both ways / gives, other hypotheses

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3. gives / fails / both ways / gives, other hypotheses

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4. gives / fails / both ways / gives, other hypotheses

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5. gives / fails / both ways / gives, other hypotheses

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13. gives / fails / both ways / gives, other hypotheses

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14. gives / fails / both ways / gives, other hypotheses

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15. gives / fails / both ways / gives, other hypotheses

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16. gives / fails / both ways / gives, other hypotheses

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One last question

Which single arrow carries the most of this chapter? Say why in one sentence. There is more than one defensible answer, and the argument is the useful part.

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